Does the orbital velocity of the pair predict the ejected body? A test against the mass ratios

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We test whether the comparison between the orbital velocity of each pair and the arrival velocity of the third body predicts which body will be ejected in the disruption of three-body systems, and whether this information adds anything to the masses. In 4,200 regularized planar integrations, with fixed mass groups and with masses drawn at random between 1 and 10, we compare the lightest body predictor, two predictors based on the velocity ratio, and a conditional logit model that combines masses and velocities, evaluated by cross-validation. With equal masses, the velocity ratio carries a weak signal, with 34% to 47% success against 33% for chance. With unequal masses, velocity alone loses to the masses, and the combined model only outperforms them when two candidates have the same mass, in which case velocity breaks the tie. With random masses there is no gain. The fit recovers an ejection probability approximately proportional to the mass raised to the power -2.4, close to the known law with exponent -3.

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I. INTRODUCTION

The previous note in the series [6] showed that, in the disruption of three-body systems with unequal masses, the rule "the pair of the two heaviest remains" predicts the outcome better than any energy signal from the trajectories. This leaves open the question of the role of velocity. In the theory of encounters between binaries and single bodies, the outcome depends on the comparison between the orbital velocity of the binary and the arrival velocity of the intruder: hard binaries, which orbit rapidly relative to a slow intruder, tend to tighten and to fling it out, and soft binaries tend to be disrupted [1, 2].

This note tests whether this velocity comparison predicts which body will be ejected and, above all, whether it adds information to the masses. The result is that velocity carries a weak signal, useful only when the masses do not decide.

II. METHOD

A. Simulations

We use the same regularized integrator [4, 5], the same ejection criteria, and the same two ensembles of initial conditions as in the previous note [6]: start from rest and virial ratio 0.5, in planar simulations with $G = 1$. We performed 300 runs per ensemble for the mass groups (1, 1, 1), (1, 1, 3), (1, 1, 10), (3, 3, 1), and (10, 10, 1), with the same seeds as in the previous note, and 600 runs per ensemble with independently drawn masses, log-uniform between 1 and 10, for a total of 4,200 runs. We analyzed the runs with an ejection preceded by at least one pericenter and a relative energy error below 10^{-4}, between 245 and 597 per row.

B. Velocity ratio

For each candidate body k, we consider the pair formed by the other two, i and j, and define

$$\rho_k = \frac{v_{\mathrm{orb}}}{V_k}, \qquad v_{\mathrm{orb}}^2 = \frac{2\,|E_{ij}|}{\mu_{ij}}, \tag{1}$$

where $E_{ij} < 0$ is the energy of the pair, μ_{ij} its reduced mass, and V_k the velocity of k relative to the center of mass of the pair; if the pair is not bound, $\rho_k = 0$. Large values of ρ_k correspond to a pair that orbits rapidly relative to a slow third body, the case in which the ejection of k is expected.

C. Predictors

We compare, at fractions $f = 0.25$, 0.5, 0.75, and 0.9 of the time to ejection: (i) masses: the lightest body is ejected, with ties counted as a fractional success; (ii) instantaneous velocity: the k with the largest ρ_k at the instant $f\,t_{\mathrm{ej}}$ is ejected; (iii) mean velocity: the k with the largest time average of $\ln\rho_k$ over the interval $[0, f\,t_{\mathrm{ej}}]$ is ejected, with ρ_k bounded from below by 0.01; (iv) combined model: a conditional logit, in which the probability of ejection of k is proportional to $\exp(\beta_1 \ln m_k + \beta_2 \ln\rho_k + \beta_3 \langle\ln\rho_k\rangle)$, fitted separately in each ensemble, with all mass groups, and evaluated by fivefold cross-validation.

III. RESULTS

Table I summarizes the success rates at $f = 0.5$, and Fig. 1 compares them for the start from rest. The results are qualitatively the same at the other time fractions.

Table I. Success rate for the ejected body at $f = 0.5$, by mass group and by ensemble (R: start from rest; V: virial). Chance: 0.33. The binomial error is about three percentage points.

Masses	Ens.	Runs	Lightest	Instantaneous velocity	Mean velocity	Combined model
1, 1, 1	R	295	0.33	0.34	0.37	0.37
1, 1, 1	V	261	0.33	0.47	0.43	0.43
1, 1, 3	R	299	0.49	0.51	0.56	0.56
1, 1, 3	V	264	0.49	0.51	0.50	0.54
1, 1, 10	R	296	0.50	0.59	0.60	0.58
1, 1, 10	V	254	0.50	0.63	0.61	0.66
3, 3, 1	R	298	0.94	0.60	0.70	0.94
3, 3, 1	V	256	0.91	0.62	0.69	0.91
10, 10, 1	R	299	0.98	0.56	0.72	0.98
10, 10, 1	V	245	0.98	0.74	0.82	0.98
random	R	597	0.71	0.54	0.57	0.72
random	V	526	0.76	0.54	0.57	0.76

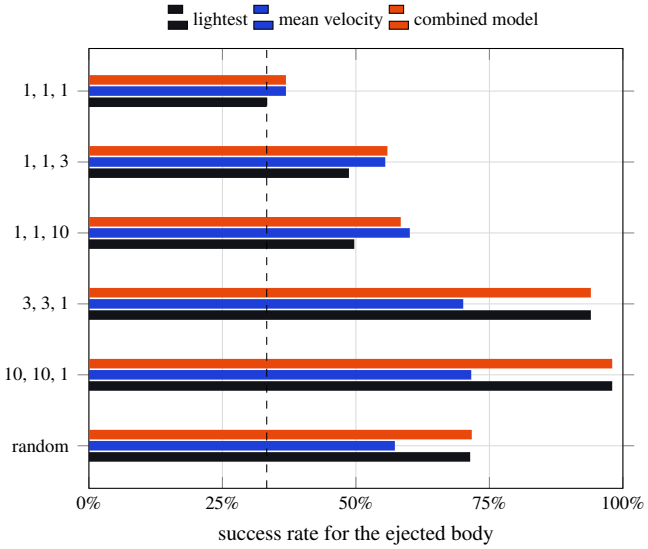


Fig. 1. Success rate for the ejected body for the start from rest, at $f = 0.5$: lightest body predictor, mean velocity predictor, and combined model. The dashed line indicates chance.

Three patterns stand out. With equal masses, the velocity ratio carries a weak signal: from 34% to 47% success, against 33% for chance, stronger in the virial ensemble. With unequal masses and one clearly lightest body, velocity alone loses to the masses by a wide margin, for example 60% to 70% against 91% to 94% in the (3, 3, 1) group. And the combined model only outperforms the mass predictor when the masses are tied, that is, when there are two equal light bodies: in the (1, 1, 3) and (1, 1, 10) groups, the success rate rises from about 50% to 54% to 66%, because velocity then breaks the tie between the two candidates. With random masses, the most general case, the combined model does not improve on the mass predictor: 72% and 76%, against 71% and 76% for the masses alone.

The fitted coefficients confirm this reading. In both ensembles, $\beta_1 \approx -2.4$, whereas β_2 and β_3 lie between 0.02 and 0.11. The model therefore assigns an ejection probability approximately proportional to $m^{-2.4}$, close to the approximate m^{-3} law [3], and almost no weight to velocity.

IV. DISCUSSION

The absence of gain with unequal masses has a simple explanation. The ratio ρ_k already depends on the masses: with one light body, the pair formed by the two heavy ones tends to be the most bound and the fastest, and the arrival velocity of the light body reflects the same imbalance. The velocity information is largely redundant with the mass information. When the masses are tied, this redundancy disappears, and what remains of the velocity is useful to break the tie.

The weak signal in the equal-mass case is consistent with the result of the previous note [6]: in the chaotic regime, the configuration changes at each encounter, and a measurement taken before the last close encounter says little about it. The theory of encounters with hard and soft binaries [1, 2] describes well the mean outcome of an isolated encounter, but, in a chaotic sequence of encounters, each one erases part of the information from the previous one.

The main limitations are the following. The simulations are planar, and the start from rest has zero total angular momentum. The ratio ρ_k is one choice among several possible ones; measures restricted to the closest encounters, or that take into account the geometry of the encounter, could carry more information. The combined model is linear in the logarithms and does not explore interactions between the variables. With 245 to 597 runs per row, the binomial error is two to three percentage points.

V. CONCLUSIONS

The comparison between the orbital velocity of the pair and the arrival velocity of the third body does not replace the masses as a predictor of the ejected body and adds almost no information to them. It works as a tiebreaker when two candidates have the same mass and gives a weak signal when the three masses are equal. As a by-product, the fit independently recovers an ejection law close to m^{-3}.

TRANSPARENCY AND USE OF ARTIFICIAL INTELLIGENCE

The author is a lawyer and not a physicist. This work results from independent research, conducted with the assistance of artificial intelligence (a language model) in the literature searches, the programming of the simulations, the analysis of the results, the writing, and the translation. The author formulated the questions, followed the experiments, and is responsible for the published content.

DATA AND CODE AVAILABILITY

All numbers were obtained with scripts published at https://fisica-revisitada.vercel.app/en/code/, which run on Node.js without external dependencies and allow the results to be reproduced.

CONFLICT OF INTEREST

The author declares no conflicts of interest. This work received no funding. This text has not undergone peer review and should be read as a working note. Corrections are recorded in the note itself, with the date.

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